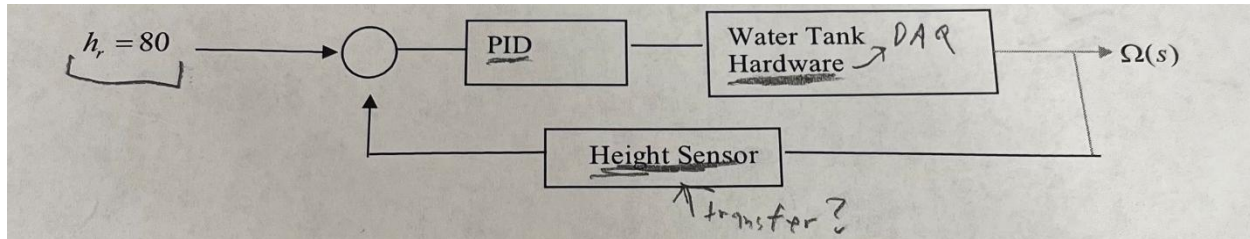


Introduction

This lab utilized a LabVIEW VI connected to a DAQ connected to a motor controlled by a PCB. The motor will adjust how much water is flowing to the water tank from the reservoir based on the measured tank level (MTL) and how fast the MTL is decreasing.



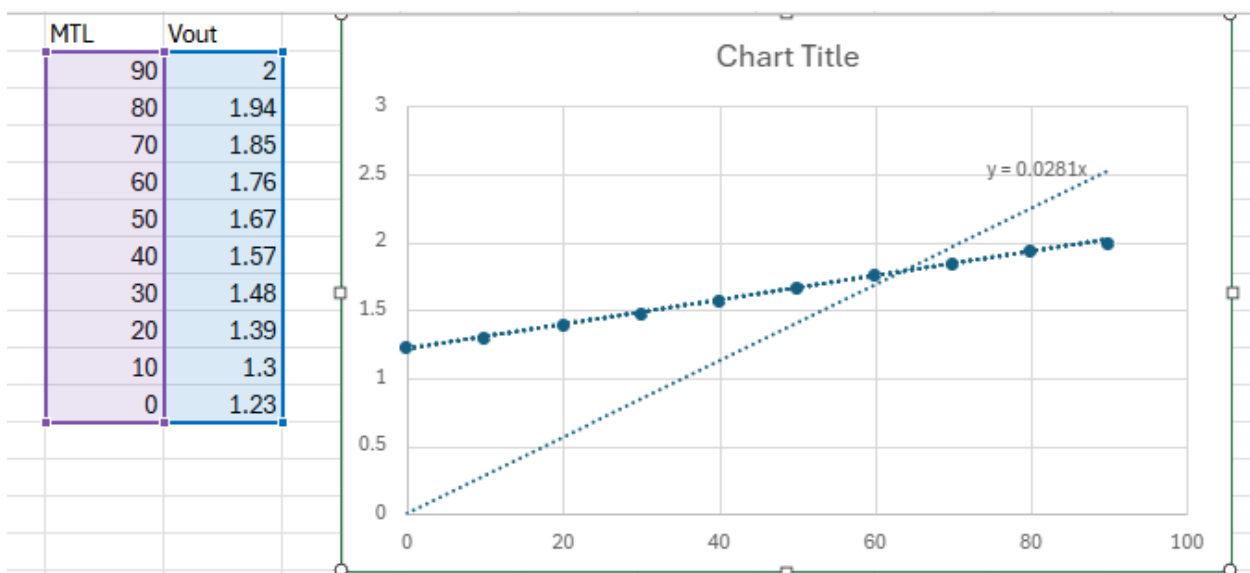
The control system for this experiment involved the desired tank level (DTL) as an input the user set, the transfer function of the dynamics of the physical system (motor, etc.), and the output of the sensor voltage that will talk to a transfer function to control the MTL. The model for the system is:



Ideally, whatever the DTL input is the system will adjust the motor speed to adjust the flow rate to keep the MTL around the DTL.

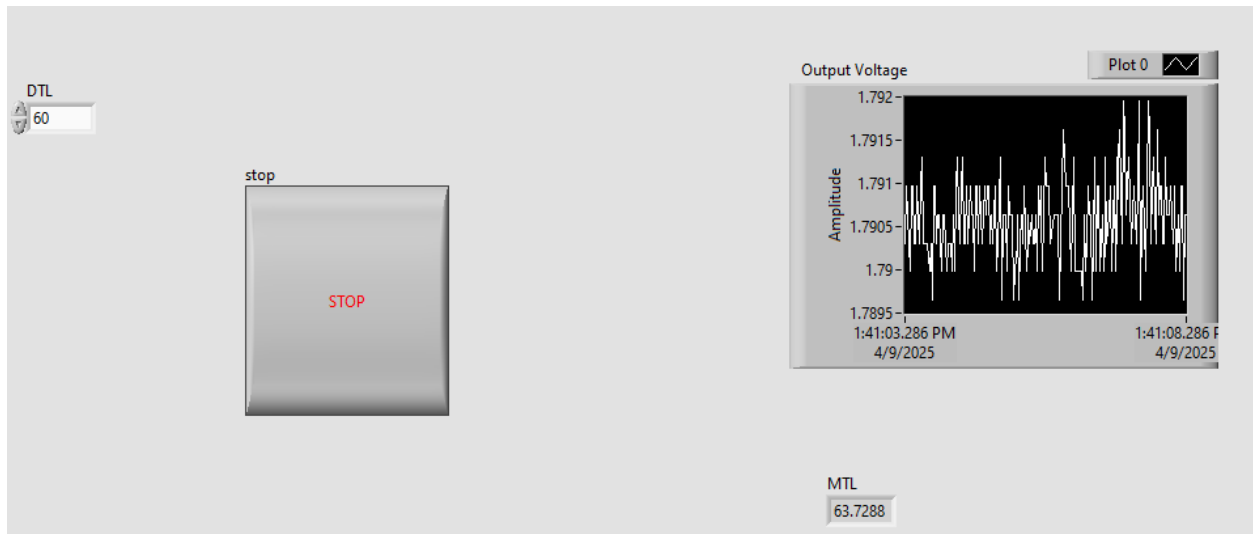
The transfer function of the water level vs. the sensor's output voltage was calculated previously in Lab 7. My transfer function was giving me bad data, so I redid the calculation of the transfer function with some new data. I realized that I needed to set the y-intercept to 0 for the line of best fit to get a proper slope estimation.

Because this chart was set up backwards and I needed to have the transfer function as the measured tank level over the output voltage, I took the inverse of this value and got 35.59 for my corrected transfer function.



Analysis

Front Panel:



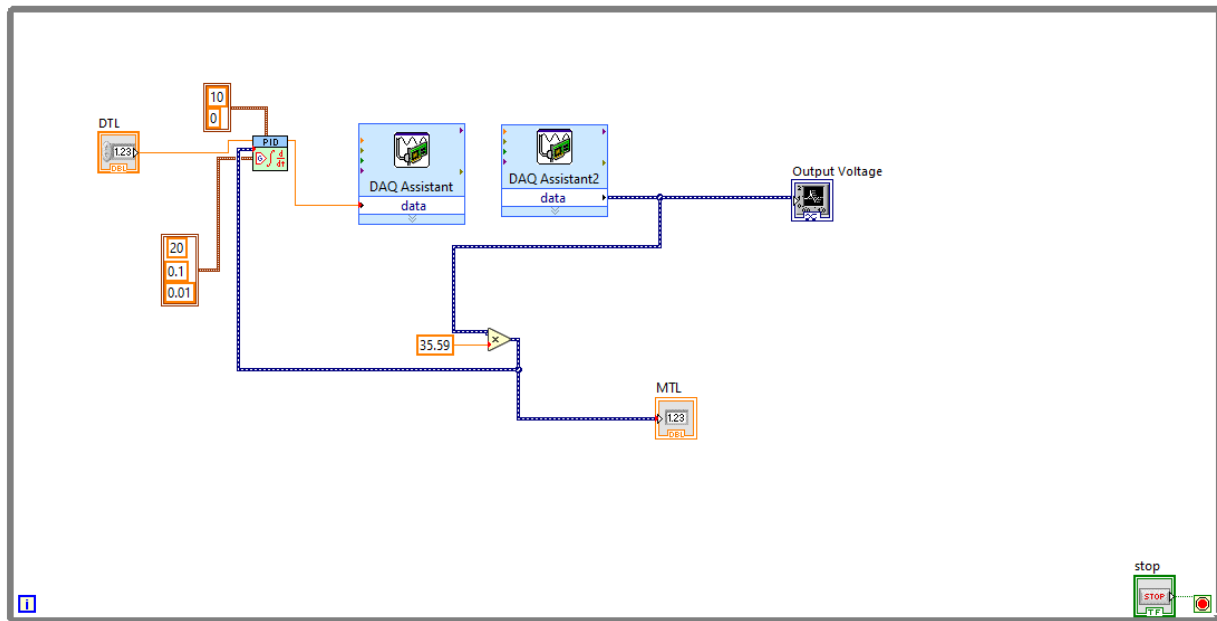
I had a numeric indicator for the desired tank level and displayed the output voltage to a waveform and a numeric display.

I had everything running in a while loop, which was doable because the transfer function of the water level sensor was a constant.

The numeric display to the front panel calculates the MTL using the transfer function multiplied by the output voltage.

This is useful for troubleshooting since it lets you know where the sensor / DAQ thinks the tank level is at. You can then use this to adjust the PID gains. We found these PID gains experimentally by comparing the MTL to the actual water tank level.

Block Diagram:



The DTL is the setpoint for the PID controller, and the PID control is connected to the transfer function / sensor.

The PID gains is connected to constants set experimentally. The output of the PID is connected to the DAQ's input. The DAQ's output feeds to the output voltage and transfer function.

The range of the PID was set between 0 and 10 to prevent the PID from outputting bad values.

Conclusion

This lab was really cool for seeing a practical application of what we have been learning.

It was fairly straightforward to implement the design, and the only real hard part was adjusting the PID values since we had to solve those experimentally rather than solving it mathematically. Solving the PID values mathematically would be part of the next course Control Systems Design.

We learned to apply a PID controller in LabVIEW to control a system, and applied it in lab.